

Appendix E: Collider Input Dataset Notes

Summary file for repository documentation

This document records the collider input dataset used in the thesis. The table is intentionally compact and focused on the results needed for the VBF-first reinterpretation. It is not intended to be an exhaustive survey of every invisible-Higgs search in the literature.

Experiment	Year	Channel / analysis	sqrt(s) TeV	Luminosity fb ⁻¹	Observed BR _{inv}	Expected BR _{inv}	Uncertainty treatment	Role in thesis	Ref
CMS	2014	VBF + ZH combined	8	19.5	0.65	0.49	95% CL; expected limit retained as sensitivity reference; detailed bands not tabulated in the working dataset	Early comparison point	[7]
CMS	2019	VBF-only	13	35.9	0.33	0.25	95% CL; expected limit retained as sensitivity reference	Run-2 VBF input	[8]
CMS	2022	VBF-only	13	101	0.18	0.10	95% CL; expected limit retained as sensitivity reference	Stronger CMS VBF input	[9]
ATLAS	2022	VBF-only	13	139	0.145	0.103	95% CL; expected +/-1 sigma and +/-2 sigma ranges available and propagated in Figures 5.1 and 5.2	Main ATLAS VBF input	[10]
ATLAS	2023	Combined invisible-Higgs searches	13	139	0.107	0.077	95% CL; expected limit retained as overall sensitivity reference	Overall reference bound	[11]

Selection logic: The selected inputs were chosen because they provide a clear historical and analytical path from early VBF-sensitive invisible-Higgs limits to stronger Run-2 VBF and combined results. The dataset supports a focused VBF-first reinterpretation rather than a full review of all Higgs production channels.

Use in analysis: Observed limits are used for the main exclusion mapping. Expected limits are retained as sensitivity references. For the ATLAS 2022 VBF input, the expected +/-1 sigma and +/-2 sigma ranges are propagated through the width and coupling translations to produce the shaded bands in the final coupling plots.

Relation to repository datasets: The numerical values in this document correspond to datasets/collider_limits.csv and datasets/atlas2022_vbf_uncertainty.csv.